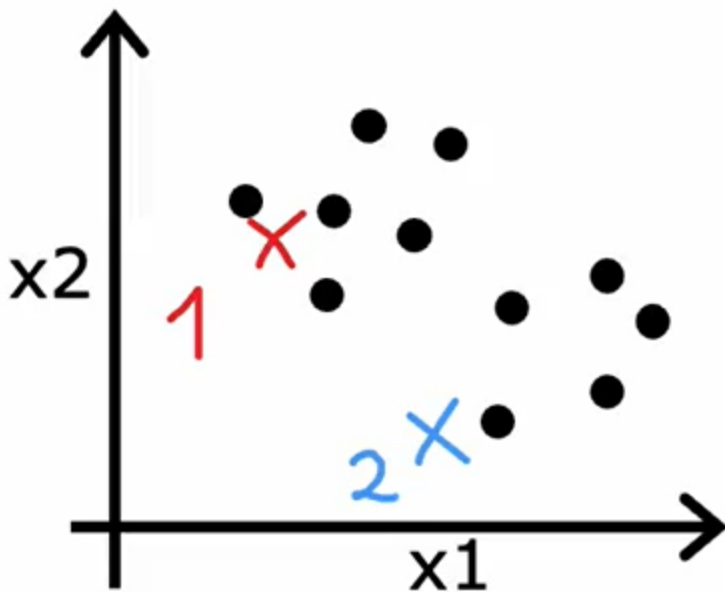


- Cluster a group of data which are similar to each other

K-Means Algorithm

Steps :

1. Randomly assign K number of Cluster Centroids
2. Calculate the distances of each datapoints from the centroid
3. Move the Centroid to get less total distance
4. At the end it find a patch of data points which are together
5. If the centroid it not assigned to any things the reinicialise it



Optimization Objective

Cost function

$$J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K) = \frac{1}{m} \sum_{i=1}^m \|x^{(i)} - \mu_{c^{(i)}}\|^2$$

$\min_{\substack{c^{(1)}, \dots, c^{(m)} \\ \mu_1, \dots, \mu_K}} J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K)$

The diagram includes two red arrows pointing down to the terms $x^{(i)}$ and $\mu_{c^{(i)}}$ in the equation, and two blue arrows pointing from the same terms towards the right, indicating the distance calculation.

- Also called as Distortion Function

Initializing K-means

Random Initialization

- Choose $K < m$
 - K is the number of centroids and m is the no of training examples
- Randomly pick some training examples
- Do this multiple time and pick the one which gave lowest J (cost function)

Choosing the the no of Clusters (K)

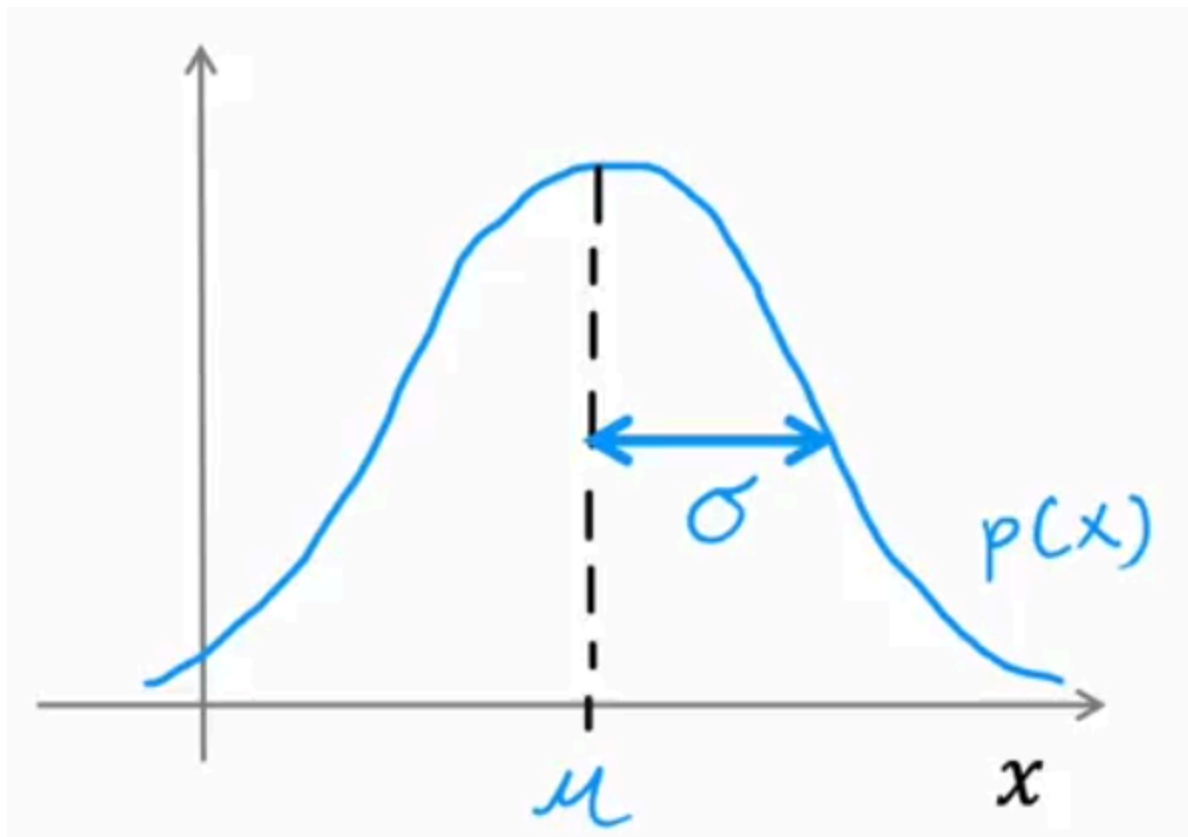
- Elbow method
 - Run K-means with a variety of values of K and compare the cost function(Distortion function) of it

Anomaly Detection

Finding Unusual Events

- Method - Density Estimation
- Used in Fraud Detection, Financial Fraud

Gaussian Distribution / Normal Distribution

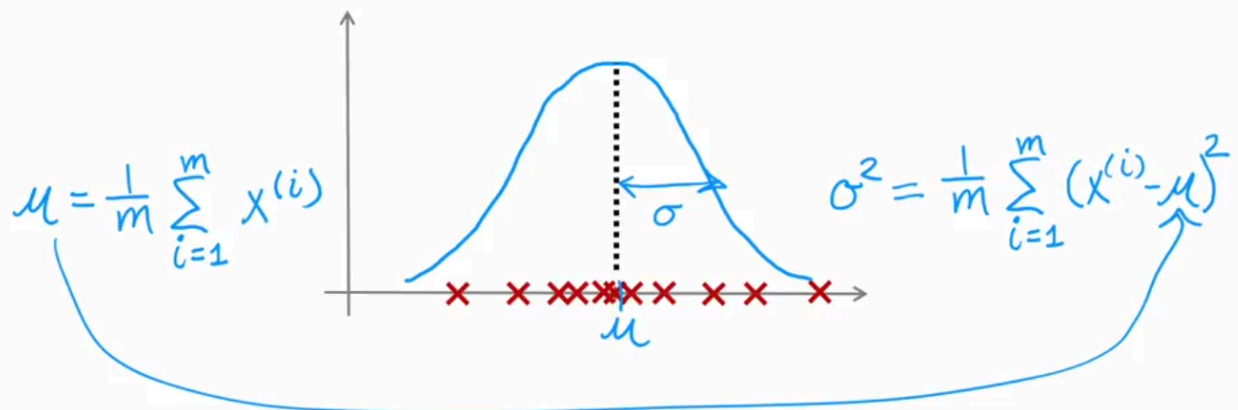


$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- The area under the curve is always equal to one

Parameter estimation

Dataset: $\{x^{(1)}, x^{(2)}, \dots, x^{(m)}\}$



$$p(x) = \prod_{j=1}^n p(x_j; \mu_j, \sigma_j^2) = \prod_{j=1}^n \frac{1}{\sqrt{2\pi}\sigma_j} \exp\left(-\frac{(x_j - \mu_j)^2}{2\sigma_j^2}\right)$$

Real Number Evaluation

- Include a few anomalous examples with labels
-

Anomaly Detection	Supervised Learning
Very small number of positive examples	Larger number of positive examples
Many different types of anomalies (Future Anomalies may not look like the trained examples)	Future examples are similar to the data trained from
Ex : Fraud Detection	Ex : Email Spam Classification

Choosing features to use

- Use Gaussian features

